



Point-Source Neutrino Search: Analysis of NGC 1068 with IceCube Data

Master's Degree in Physics

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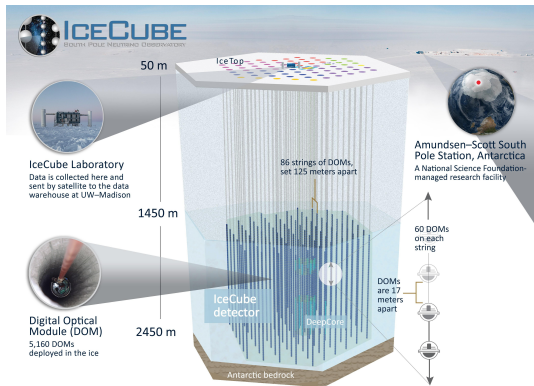
July 21, 2026



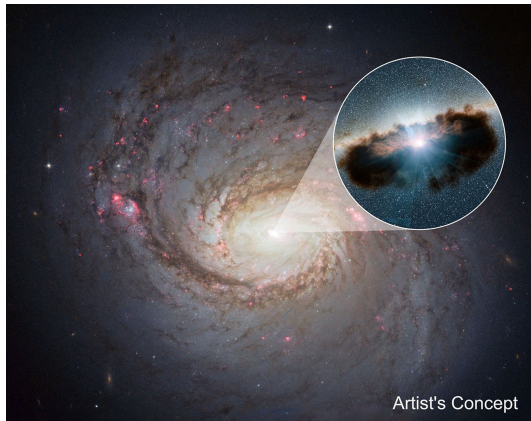
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The IceCube Observatory and NGC 1068



Source: icecube.wisc.edu



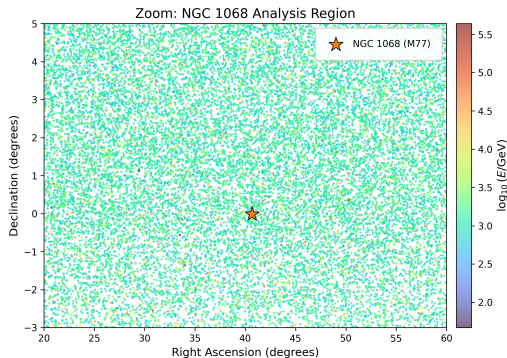
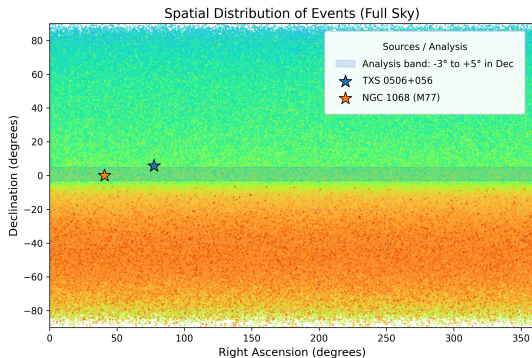
Source: science.nasa.gov



Dataset Selection & Spatial Cuts

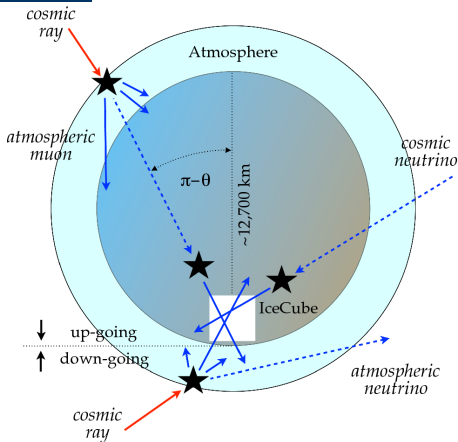
The **Tracks dataset** provides the excellent angular resolution ($\sim 1^\circ - 2^\circ$) required for point-source astronomy.

The analysis is strictly confined to a narrow **declination band** ($\delta \in [-3^\circ, +5^\circ]$).





Data-Driven Background Estimation via RA Scrambling



Source: link.springer.com

Analysis Region

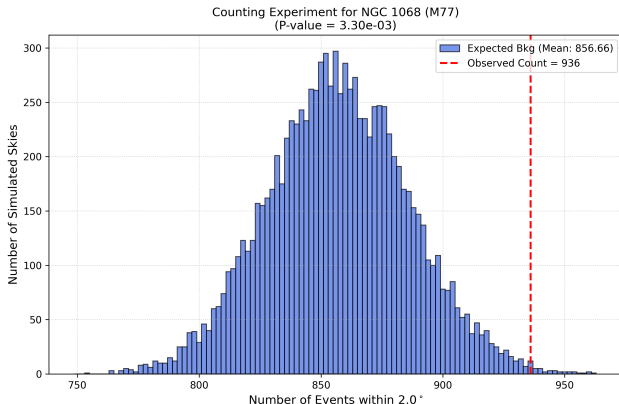
- **Target:** NGC 1068 ($\delta \approx -0.01^\circ$)
- **Band Cut:** $\delta \in [-3^\circ, +5^\circ]$
- Assumes locally uniform detector acceptance.

Background Modeling

- **Technique:** Right Ascension (RA) Scrambling
- **Process:** Fix δ_i , randomize α_i
- Data-driven baseline intrinsically accounting for exposure dependencies.



First signal search for NGC 1068



Counting Analysis Results

Parameter	Value
Simulated Skies	10^4
Observed Events	936
Exp. Background	856.7 ± 29.1
Excess Events	$+79.3$
<i>p</i> -value	3.3×10^{-3}
Significance	2.72σ

Note: A simple binned circular search provides a solid baseline but does not exploit other important information.



Unbinned Spatial Likelihood

Mathematical Framework

$$\mathcal{L}(n_s) = \prod_{i=1}^N \left[\frac{n_s}{N} \mathcal{S}_i + \left(1 - \frac{n_s}{N} \right) \mathcal{B}_i \right]$$

- N : Total events in the declination band.
- n_s : Number of signal events (*free parameter*).

Probability Density Functions (PDFs)

Signal PDF ($\mathcal{S}_i$)

$$\mathcal{S}_i = \frac{1}{2\pi\sigma_i^2} \exp\left(-\frac{\psi_i^2}{2\sigma_i^2}\right)$$

Background PDF ($\mathcal{B}_i$)

$$\mathcal{B}_i = \frac{1}{\Delta\Omega_{\text{band}}} = \frac{1}{2\pi(\sin\delta_{\text{max}} - \sin\delta_{\text{min}})}$$



Likelihood Scan and Best-Fit Results

- The Test Statistic (TS) is evaluated as the likelihood ratio against the null hypothesis:

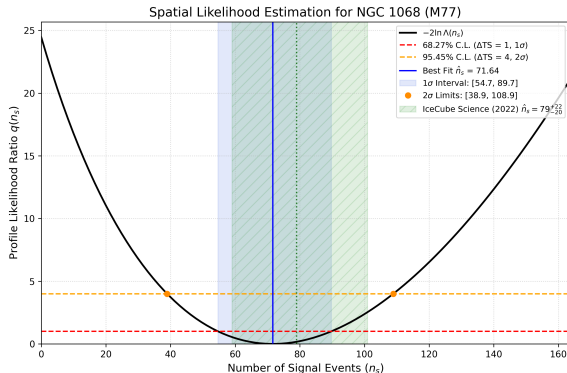
$$TS = 2 \log \left(\frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(n_s = 0)} \right)$$

- Best-fit signal:**

$$\hat{n}_s = 71.64^{+18.08}_{-16.94} \quad 68.27\% \text{ C.L.}$$

- Observed TS:** $TS_{obs} = 24.46$

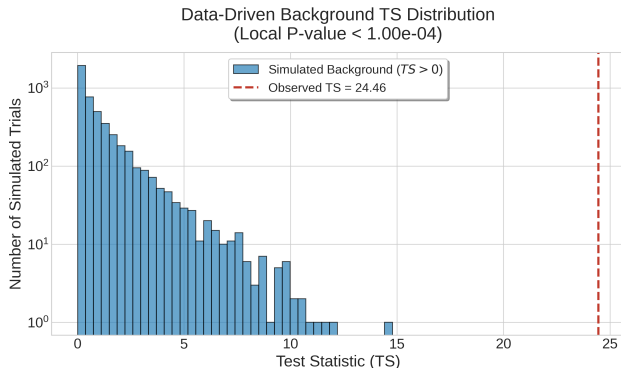
- Significance:** 4.9σ $p_{value} = 3.8 \times 10^{-7}$
(assuming the conditions of Wilks' theorem are satisfied)





Post-trial Significance Assessment via RA Scrambling

Empirical background TS distribution derived via Right Ascension scrambling ($N = 5000$) to bypass Wilks' theorem asymptotic approximations at the target declination.



Statistical Results

- INSERT background fluctuations.
- **Significance:** INSERT
- **p-value:** INSERT



Conclusions & Outlook

A Solid Spatial-Only Baseline

- ✓ Successfully implemented a data-driven, unbinned spatial likelihood.
- ✓ Identified a clear neutrino excess spatially consistent with NGC 1068, yielding a robust, model-independent empirical significance.

Current Limitations & Future Directions

- ! **Energy Discrimination:** Neglecting lepton energy discards crucial separation power between hard astrophysical spectra and atmospheric backgrounds.
- ! **Computational Complexity:** True neutrino energy unfolding requires demanding Monte Carlo simulations and advanced Instrument Response Functions (IRFs).
- ! **Outlook:** Integrate energy-dependent PDFs to fully unlock the dataset's potential, and benchmark this custom pipeline against the official IceCube **SkyLLH** framework.



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Thank you for listening!
Any questions?